Thermodynamics EMCH 290

Problem #1

$$PV = mRT$$

$$R = \frac{1545\ ft * lbf}{28.96\ lb * °R}$$

$$T_L = 40°F = 499.67°R;\ T_R = 200°F = 659.67°R$$

$$m_L = \frac{p_L * V_L}{R * T_L} = \frac{14.7\frac{lb}{in^2} * 144\frac{in^2}{ft^2} * 12ft^3}{\frac{1545\ ft * lbf}{28.96\ lb * °R} * 500°R} = \frac{7662816}{8046875}$$

$$m_R = \frac{p_R * V_R}{R * T_R} = \frac{50\frac{lb}{in^2} * 144\frac{in^2}{ft^2} * 10ft^3}{\frac{1545\ ft * lbf}{28.96\ lb * °R} * 660°R} = \frac{11584}{5665}$$

$$P = \frac{mRT}{V} \to \frac{\left(\frac{11584}{5665} + \frac{7662816}{8046875}\right) lb * \frac{1545\ ft * lbf}{28.96\ lb * °R} * (149.3 + 459.67)°R}{22ft^3} * \frac{1}{144}\frac{ft^2}{in^2} = 30.735\frac{lb}{in^2}$$

$$\sigma = m_L\left[s(T_F) - s(T_L) - R * \ln\left(\frac{p_F}{p_L}\right)\right] + m_R\left[s(T_F) - s(T_R) - R * \ln\left(\frac{p_F}{p_R}\right)\right]$$

$$s(T_F) = 0.629616; s(T_L) = 0.58233; s(T_R) = 0.64902;$$

$$\sigma = \frac{7662816}{8046875} * \left(0.629616 - 0.58233 - \frac{1545\ ft * lbf}{28.96\ lb * °R} * \frac{1\ Btu}{778\ ft * lbf} * \ln\left(\frac{30.735}{14.7}\right)\right) + \frac{11584}{5665} * \left(0.629616 - 0.64902 - \frac{1545\ ft * lbf}{28.96\ lb * °R} * \frac{1\ Btu}{778\ ft * lbf} * \ln\left(\frac{30.735}{50}\right)\right) = 0.02542\frac{Btu}{°R}$$

PROBLEM #2

a)

$$\frac{\sigma}{m} = -\frac{\left(\frac{Q}{m}\right)}{T_B} + (s_2 - s_1)$$

$$0 = \dot{Q} - \dot{W} + \dot{m}\left(h_1 - h_2 + \frac{V_1^2 - V_2^2}{2}\right) \to 0 = \frac{\dot{Q}}{\dot{m}} - \frac{\dot{W}}{\dot{m}} + \left(h_1 - h_2 + \frac{V_1^2 - V_2^2}{2}\right) \to$$

$$\frac{\dot{Q}}{\dot{m}} = \frac{\dot{W}}{\dot{m}} - \left(h_1 - h_2 + \frac{V_1^2 - V_2^2}{2}\right)$$

b)

$$h_1 = 3231.7\frac{kJ}{kg}\,; h_2 = 2675.6\frac{kJ}{kg}$$

c)

$$s_1 = 6.9235\ \frac{kJ}{kg} * {}^\circ K; s_2 = 7.3542\ \frac{kJ}{kg} * {}^\circ K;$$

d)

$$\frac{\dot{Q}}{\dot{m}} = \frac{\dot{W}}{\dot{m}} - \left(h_1 - h_2 + \frac{V_1^2 - V_2^2}{2}\right) \to 540\frac{kJ}{kg} - \left(3231.7 - 2675.6 + \frac{160^2 - 100^2}{2} * \frac{1}{1000}\right) = -23.9\frac{kJ}{kg}$$

e)

$$\frac{\sigma}{m} = -\frac{\left(\frac{Q}{m}\right)}{T_B} + (s_2 - s_1) \to \frac{\sigma}{m} = -\frac{(-23.9)}{350} + (7.3542 - 6.9235) = 0.4989\frac{kJ}{kg} * {}^\circ K$$